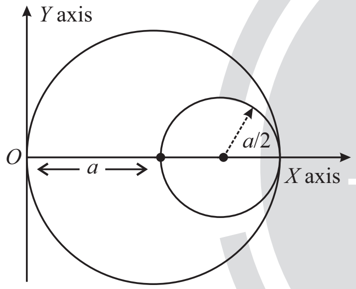
Motion of system of particle and COM

Single Correct Type Questions

1. A circular hole of radius $\left(\frac{a}{2}\right)$ is cut of a circular disc of radius 'a' as shown in figure. The centroid of the remaining circular portion with respect to point 'O' will be:

[24 Feb, 2021 (Shift-II)]



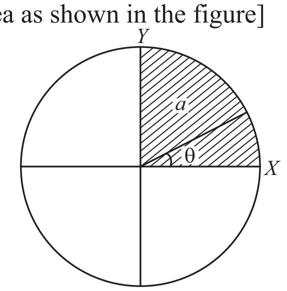
(*a*) $\frac{5}{6}a$ (*b*) $\frac{1}{6}a$

(*c*) $\frac{10}{11}$a (*d*) $\frac{2}{3}a$

2. The disc of mass M with uniform surface mass density σ is shown in the figure. The centre of mass of the quarter disc (the shaded area) is at the position $\frac{x}{3}\frac{a}{\pi}, \frac{x}{3}\frac{a}{\pi}$ where x is __(Round off to the Nearest Integer)

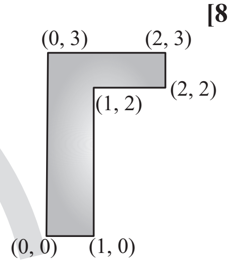
[17 March, 2021 (Shift-II)]

[a is an area as shown in the figure]



3. The coordinates of centre of mass of a uniform flag shaped lamina (thin flat plate) of mass 4kg. (The coordinates of the same are shown in figure) are:

[8 Jan, 2020 (Shift-I)]

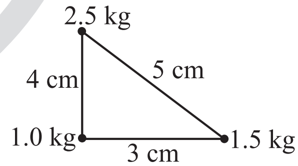


(*a*) 1.25 m, 1.50 m) (*b*) (0.75 m, 0.75 m)

(*c*) (0.75 m, 1.75 m) (*d*) (1 m, 1.75 m)

4. Three point particles of masses 10 kg, 1.5 kg and 2.5 kg are placed at three corners of a right angle triangle of sides 4.0 cm, 3.0 cm and 5.0 cm as shown in the figure. The center of mass of the system is at a point:

[7 Jan, 2020 (Shift-I)]



(*a*) 0.6 cm right and 2.0 cm above 1kg mass

(*b*) 2.0 cm right and 0.9 cm above 1 kg mass

(*c*) 0.9 cm right and 2.0 cm above 1kg mass

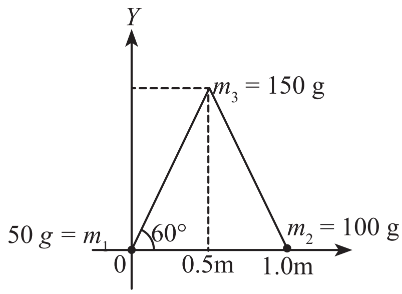
(*d*) 1.5 cm right and 1.2 cm above 1kg mass

5. A rod of length L has non-uniform linear mass density given by $\rho(x) = a + b\left(\frac{x}{L}\right)^2$ where a and b are constants and $0 \le x \le L$. The value of x for the centre of mass of the rod is at: **[9 Jan, 2020 (Shift-II)]**

(*a*) $\frac{4}{3}\left(\frac{a+b}{2a+3b}\right)L$ (*b*) $\frac{3}{4}\left(\frac{2a+b}{3a+b}\right)L$

(*c*) $\frac{3}{2}\left(\frac{2a+b}{3a+b}\right)L$ (*d*) $\frac{3}{2}\left(\frac{a+b}{2a+b}\right)L$

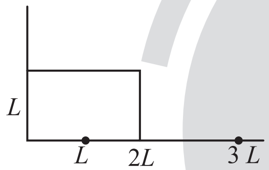
6. Three particles of masses 50 g, 100g and 150g are placed at the vertices of an equilateral triangle of side 1 m (as shown in the figure). The (x, y) coordinates of the centre of mass will be : **[12 April, 2019 (Shift-II)]**



(a) $\left(\frac{\sqrt{3}}{7}m, \frac{7}{12}m\right)$ (b) $\left(\frac{7}{12}m, \frac{\sqrt{3}}{8}m\right)$

(c) $\left(\frac{\sqrt{3}}{4}m, \frac{5}{12}m\right)$ (d) $\left(\frac{7}{12}m, \frac{\sqrt{3}}{4}m\right)$

7. The position vector of the centre of mass $\vec{r}$ cm of an asymmetric uniform bar of negligible area of cross-section as shown in figure is: **[12 Jan, 2019 (Shift-I)]**



(a) $\vec{r}\text{ (cm)} = \frac{13}{8}L\hat{x} + \frac{5}{8}L\hat{y}$

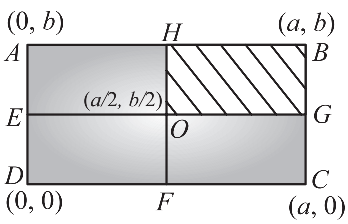
(b) $\vec{r}\text{ (cm)} = \frac{5}{8}L\hat{x} + \frac{13}{8}L\hat{y}$

(c) $\vec{r}\text{ (cm)} = \frac{3}{8}L\hat{x} + \frac{11}{8}L\hat{y}$

(d) $\vec{r}\text{ (cm)} = \frac{11}{8}L\hat{x} + \frac{3}{8}L\hat{y}$

8. A uniform rectangular thin sheet ABCD of mass M has length a and breadth b, as shown in the figure. If the shaded portion HBGO is cut off, the coordinates of the centre of mass of the remaining portion will be:

[8 April, 2019 (Shift-II)]



(a) $\left(\frac{2a}{3}, \frac{2b}{3}\right)$ (b) $\left(\frac{5a}{3}, \frac{5b}{3}\right)$

(c) $\left(\frac{3a}{4}, \frac{3b}{4}\right)$ (d) $\left(\frac{5a}{12}, \frac{5b}{12}\right)$

9. The initial mass of a rocket is 1000 kg. Calculate at what rate the fuel should be burnt so that the rocket is given an acceleration of 20 ms^{-2}. The gases come out at a relative speed of 500 ms^{-1} with respect to the rocket :

[Use g = 10 m/s^2] **[26 Aug, 2021 (Shift-I)]**

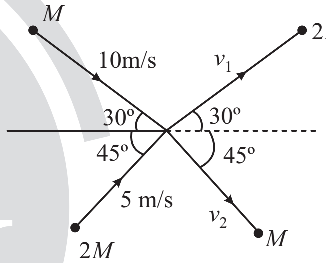
(a) 500 kgs^{-1}

(b) 10 kgs^{-1}

(c) 6.0 × 10^2 kg s^{-1}

(d) 60 kg s^{-1}

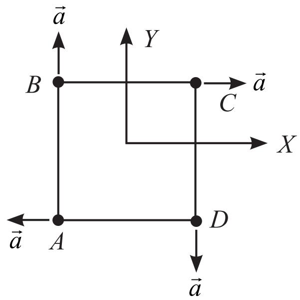
10. Two particles, of masses M and 2M, moving as shown, with speeds of 10 m/s and 5 m/s collide elastically at the origin. After the collision, they move along the indicated directions with speeds v_1 and v_2 respectively. The value of v_1 and v_2 are nearly: **[10 April, 2019 (Shift-I)]**



(a) 3.2 m/s and 6.3 m/s (b) 3.2 m/s and 12.6 m/s

(c) 6.5 m/s and 6.3 m/s (d) 6.5 m/s and 3.2 m/s

11. Four particles A, B, C and D with masses $m_A = m$, $m_B = 2m$, $m_C = 3m$ and $m_D = 4$m are at the corners of a square. They have accelerations of equal magnitude with directions as shown. The acceleration of the centre of mass of the particles is: **[8 April, 2019 (Shift-I)]**



(a) $\frac{a}{5}\left(\hat{i} - \hat{j}\right)$ (b) Zero

(c) $\frac{a}{5}\left(\hat{i} \times \hat{j}\right)$ (d) $a\left(\hat{i} + \hat{j}\right)$

12. A particle of mass m moving with velocity v collides with a stationary particle of mass $2m$. After collision, they stick together and continue to move together with velocity

[10 April, 2023 Shift-1)]

(*a*) v

(*b*) $\frac{v}{2}$

(*c*) $\frac{v}{3}$

(*d*) $\frac{v}{4}$

13. Velocity (υ) and acceleration (a) in two systems of units 1 and 2 are related as $\upsilon_2 = \frac{n}{m^2}\upsilon_1$ and $a_2 = \frac{a_1}{mn}$ respectively. Here m and n are constants. The relations for distance and time in two system respectively are:

[28 June, 2022 (Shift-II)]

(*a*) $\frac{n^3}{m^3}L_1 = L_2$ and $\frac{n^2}{m}T_1 = T_2$

(*b*) $L_1 = \frac{n^4}{m^2}L_2$ and $T_1 = \frac{n^2}{m}T_2$

(*c*) $L_1 = \frac{n^2}{m}L_2$ and $T_1 = \frac{n^4}{m^2}T_2$

(*d*) $\frac{n^2}{m}L_1 = L_2$ and $\frac{n^4}{m^2}T_1 = T_2$

14. Two blocks of masses 10 kg and 30 kg are placed on the same straight line with coordinates (0, 0) and (x, 0) respectively. The block of 10 kg is moved on the same line through a distance of 6 cm towards the other block. The distance through which the block of 30 kg must be moved to keep the position of centre of mass of the system unchanged is

[27 June, 2022 (Shift-I)]

(*a*) 4 cm towards the 10 kg block

(*b*) 2 cm away from the 10 kg block

(*c*) 2 cm towards the 10 kg block

(*d*) 4 cm away from the 10 kg block

15. Two bodies of mass 1 kg and 3 kg have position vectors $\hat{i} + 2\hat{j} + \hat{k}$ and $-3\hat{i} - 2\hat{j} + \hat{k}$ respectively. The magnitude of position vector of centre of mass of this system will be similar to the magnitude of vector:

[29 July, 2022 (Shift-I)]

(*a*) $\hat{i} - 2\hat{j} + \hat{k}$

(*b*) $-3\hat{i} - 2\hat{j} + \hat{k}$

(*c*) $-2\hat{i} + 2\hat{k}$

(*d*) $-2\hat{i} - \hat{j} + \hat{k}$

16. A block moving horizontally on a smooth surface with a speed of 40 m/s splits into two parts with masses in the ratio of 1:2. If the smaller part moves at 60 m/s in the same direction, then the fractional change in kinetic energy is:

[31 Aug, 2021 (Shift-II)]

(*a*) 2/3

(*b*) 1/3

(*c*) 1/4

(*d*) 1/8

17. A particle of mass m is moving with speed 2 v collides with a mass 2m moving with speed v in the same direction. After collision, the first mass is stopped completely while the second one splits into two particles each of mass m, which move at angle 45° with respect to the original direction. The speed of each of the moving particle will be:

[9 April, 2019 (Shift-II)]

(*a*) $v/(2\sqrt{2})$

(*b*) $2\sqrt{2}v$

(*c*) $\sqrt{2}v$

(*d*) $v/\sqrt{2}$

18. A bullet of '4 g' mass is fired from a gun of mass 4 kg. If the bullet moves with the muzzle speed of 50 ms^{-1}, the impulse imparted to the gun velocity of recoil of gun are:

[22 July, 2021 (Shift-II)]

(*a*) 0.4kg ms^{-1}, 0.1ms^{-1}

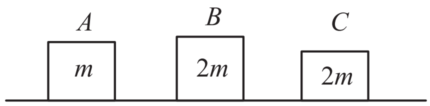
(*b*) 0.2 kg ms^{-1}, 0.05 ms^{-1}

(*c*) 0.4 kg ms^{-1},0.05 ms^{-1}

(*d*) 0.2 kg ms^{-1},0.1ms^{-1}

19. Three objects A, B and C are kept in a straight line on a frictionless horizontal surface. The masses of A, B and C are m, 2 m and 2 m respectively. A moves towards B with a speed of 9 m/s and makes an elastic collision with it. Thereafter B makes a completely inelastic collision with C. All motions occur along same straight line. The final speed of C is:

[27 July, 2021 (Shift-I)]



(*a*) 4 m/s

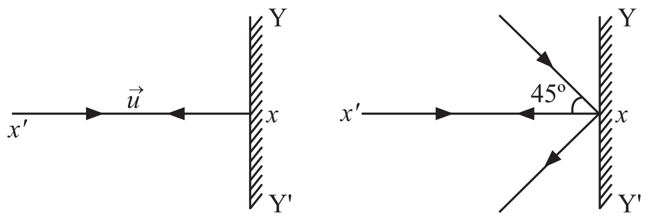
(*b*) 6 m/s

(*c*) 9 m/s

(*d*) 3 m/s

20. Two billiard balls of equal mass 30g strike a rigid wall with same speed of 108 kmph (as shown) but at different angles. If the balls get reflected with the same speed then the ratio of the magnitude of impulses imparted to ball '*a*' and ball '*b*' by the wall along '*X*' direction is:

[25 July, 2021 (Shift-I)]



Ball(a) Ball(b)

(*a*) $\sqrt{2}:1$ (*b*) $1:\sqrt{2}$

(*c*) 1 : 1 (*d*) 2 : 1

21. A particle of mass *m* is projected with a speed *u* from the ground at an angle $\theta = \pi/3$ w.r.t. horizontal (x – axis). When it has reached its maximum height, it collides completely inelastically with another particle of the same mass and velocity $u\hat{i}$. The horizontal distance covered by the combined mass before reaching the ground is:

[9 Jan, 2020 (Shift-II)]

(*a*) $\frac{3\sqrt{3}}{8}\frac{u^2}{g}$ (*b*) $2\sqrt{2}\frac{u^2}{g}$

(*c*) $\frac{5}{8}\frac{u^2}{g}$ (*d*) $\frac{3\sqrt{2}}{4}\frac{u^2}{g}$

22. A particle of mass *m* is dropped from a height *h* above the ground. At the same time another particle of the same mass is thrown vertically upwards from the ground with a speed of $\sqrt{2gh}$. If they collide head-on completely inelastically, the time taken for the combined mass to reach the ground, in units of $\sqrt{\frac{h}{g}}$ is: **[8 Jan, 2020 (Shift-II)]**

(*a*) $\frac{1}{2}$ (*b*) $\sqrt{\frac{1}{2}}$

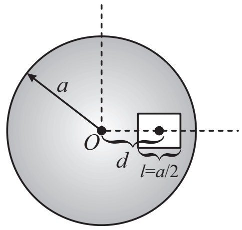
(*c*) $\sqrt{\frac{3}{4}}$ (*d*) $\sqrt{\frac{3}{2}}$

Integer Type Questions

23. The distance of centre of mass from end *A* of a one dimensional rod (*AB*) having mass density $Q = Q_0\left(1-\frac{x^2}{L^2}\right)$ kg/m and length *L* (in meter) is $\frac{3L}{\alpha}m$. The value of α is ________. (where *x* is the distance from end *A*)

[28 July, 2022 (Shift-II)]

24. A square shaped hole of side $l = \frac{a}{2}$ is carved out at a distance $d = \frac{a}{2}$ from the centre '*O*' of a uniform circular disk of radius *a*. If the distance of the centre of mass of the remaining portion from *O* is $-\frac{a}{x}$, value of *x* (to the nearest integer) is __. **[2 Sep, 2020 (Shift-II)]**

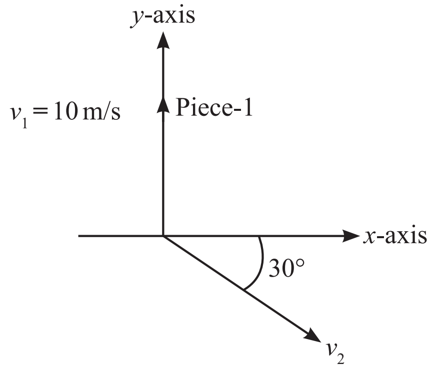


25. Three identical spheres each of mass *M* are placed at the corners of a right angled triangle with mutually perpendicular sides equal to 3 m each. Taking point of intersection of mutually perpendicular sides as origin, the magnitude of position vector of centre of mass of the system will be $\sqrt{x}$ m. The value of *x* is ________.

[25 July, 2022 (Shift-II)]

26. A ball of mass 10kg, moving with a velocity $10\sqrt{3}$m/s along, the *x*-axis, hits another ball of mass 20kg which is at rest. After the collision, first ball comes to rest while second ball disintegrates into two equal pieces. One piece starts moving along *y*–axis with a speed of 10 m/s. The second piece starts moving at an angle of 30° with respect to the *x*-axis.

The velocity of the ball moving at 30° with *x*-axis is *x* m/s. The configuration of pieces after collision is shown in the figure below.



The value of *x* to the nearest integer is ________.

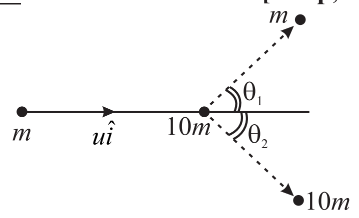
[18 March, 2021 (Shift-I)]

27. A bullet of 10 g, moving with velocity v, collides head-on with the stationary bob of a pendulum and recoils with velocity 100 m/s. The length of the pendulum is 0.5 m and mass of the bob is 1 kg. The minimum value of v = _____ m/s so that the pendulum describes a circle. (Assume the string to be inextensible and $g = 10$ m/s^2)

[27 Aug, 2021 (Shift-II)]

28. A particle of mass m is moving along the x-axis with initial velocity $u\hat{i}$. It collides elastically with a particle of mass $10m$ at rest and then moves with half its initial kinetic energy(see figure). If $\sin\theta_1 = \sqrt{n}\sin\theta_2$, then value of n is ________.

[2 Sep, 2020 (Shift-II)]



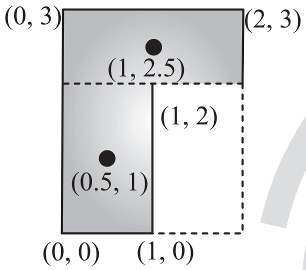
ANSWER KEY

1. (*a*)	**2.** (*c*)	**3.** (*c*)	**4.** (*b*)	**5.** (*d*)	**6.** (*a*)	**7.** (*d*)	**8.** (*d*)	**9.** (*c*)	**10.** (*a*)
11. (*c*)	**12.** (*a*)	**13.** (*c*)	**14.** (*d*)	**15.** (*d*)	**16.** (*b*)	**17.** (*b*)	**18.** (*d*)	**19.** (*a*)	**20.** (*a*)
21. (*d*)	**22.** [8]	**23.** [4]	**24.** [23]	**25.** [2]	**26.** [20]	**27.** [400]	**28.** [10]		

EXPLANATIONS

1. (*a*) $x_{\text{COM}} = \dfrac{(\sigma\pi^2 a)a - \sigma\pi\left(\frac{a}{2}\right)^2 \cdot \left(\frac{3a}{2}\right)}{\sigma\pi^2 a - \sigma\pi\left(\frac{a}{2}\right)^2} = \dfrac{a - \frac{3a}{8}}{1 - \frac{1}{4}} = \dfrac{5a}{6}$

2. (*c*)



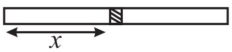
$$\vec{r}_{\text{cm}} = \frac{2\times\left(\frac{\hat{i}}{2} + \hat{j}\right) + 2\times\left(\hat{i} + \frac{5\hat{j}}{2}\right)}{2+2} \Rightarrow \vec{r}_{\text{cm}} = \frac{3}{4}\hat{i} + \frac{7}{4}\hat{j}$$

3. (*c*) From 1 kg mass,

$$x_{\text{COM}} = \frac{1(0) + 2.5(0) + 1.5(3)}{1 + 2.5 + 1.5} = \frac{4.5}{5} = 0.9\,\text{cm}$$

$$y_{\text{COM}} = \frac{1(0) + 2.5(4) + 1.5(0)}{1 + 2.5 + 1.5} = 2\,\text{cm}$$

4. (*b*)



$$x_{\text{cm}} = \frac{1}{M}\int_0^L x.dM$$

$$dM = \lambda.dx = \left(a + b\left(\frac{x}{L}\right)^2\right).dx$$

$$x_{cm} = \frac{\int x dM}{\int dM} = \frac{\int x\lambda dx}{\int \lambda dx} = \frac{\int_0^L x\left(a + \frac{bx^2}{L^2}\right)dx}{\int_0^L \left(a + \frac{bx^2}{L^2}\right)dx}$$

$$= \frac{a\left(\frac{x^2}{2}\right)_0^L + \frac{b}{L^2}\left(\frac{x^4}{4}\right)_0^L}{a(x)_0^L + \frac{b}{L^2}\left(\frac{x^3}{3}\right)_0^L}$$

$$= \frac{a\frac{L^2}{2} + b\frac{L^2}{4}}{aL + \frac{bL}{3}} = \frac{(2a+b)}{(3a+b)}\frac{L}{4}\times 3$$

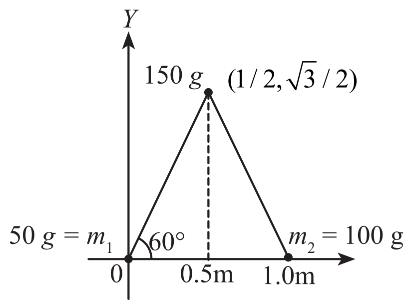
$$= \frac{3L}{4}\left(\frac{2a+b}{3a+b}\right)$$

5. (*d*) $\vec{r}_{cm} = \dfrac{m_1\vec{r}_1 + m_2\vec{r}_2 + m_3\vec{r}_3}{m_1 + m_2 + m_3}$

$$\vec{r}_{cm} = \frac{0 + 150\times\left(\frac{1}{2}i + \frac{\sqrt{3}}{2}\hat{j}\right) + 100\times\hat{i}}{300}$$

$$\vec{r}_{cm} = \frac{7}{12}\hat{i} + \frac{\sqrt{3}}{4}\hat{j}$$

$\therefore$Co-ordinate of center $= \left(\dfrac{7}{12}, \dfrac{\sqrt{3}}{4}\right)m$



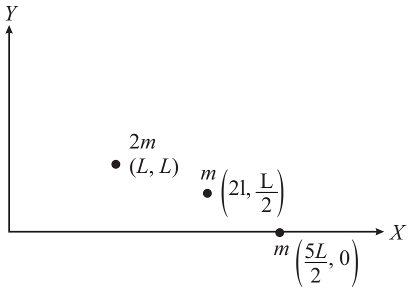
6. (*a*) Let consider the rod in three parts of masses $2m$, m and m.

$$\vec{r}_1 = L\hat{i} + L\hat{j}$$

$$\vec{r}_2 = 2L\hat{i} + \frac{L}{2}\hat{j}$$

$\vec{r}_3 = \frac{5L}{2}\hat{i}$

Three parts of rod can be considered as point masses.



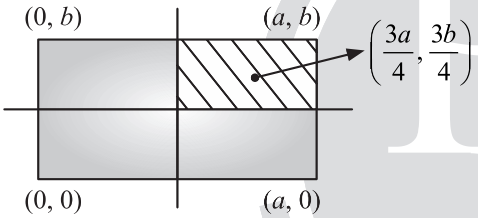
Now, position vector of COM,

$$\vec{r}_{cm} = \frac{2m\,\vec{r}_1 + m\,\vec{r}_2 + m\,\vec{r}_3}{4m}$$

$$\vec{r}_{cm} = \frac{2m\left(L\hat{i} + L\hat{j}\right) + m\left(2L\hat{i} + \frac{L}{2}\hat{j}\right) + m\left(\frac{5L}{2}\hat{i}\right)}{4m}$$

$$= \frac{13}{8}L\hat{i} + \frac{5}{8}L\hat{j}$$

7. (*d*)



Area of complete sheet $(A_1) = ab$

Area of shaded part $(A_2) = \frac{ab}{4}$

For centre of mass,

$$X_{\text{cm}} = \frac{A_1X_1 - A_2X_2}{A_1 - A_2} = \frac{ab\left(\frac{a}{2}\right) - \frac{ab}{4}\left(\frac{3a}{4}\right)}{ab - \frac{ab}{4}}$$

$$X_{\text{cm}} = \frac{5a}{12}$$

$$Y_{\text{cm}} = \frac{A_1Y_1 - A_2Y_2}{A_1 - A_2} = \frac{ab\left(\frac{a}{2}\right) - \frac{ab}{4}\left(\frac{3b}{4}\right)}{ab - \frac{ab}{4}}$$

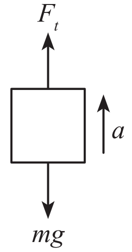
$$Y_{\text{cm}} = \frac{5b}{12}$$

So, the coordinate of centre of mass (remaining sheet)

$$(X_{\text{cm}}, Y_{\text{cm}}) \equiv \left(\frac{5a}{12}, \frac{5b}{12}\right)$$

8. (*d*) Formula of thrust, $F_t = \left(\frac{dm}{dt}\right)V_{re\ell}$

Net force = ma



$F_t - mg = ma$

$\left(\frac{dm}{dt}\right)V_{re\ell} - mg = ma$

$$500\frac{dm}{dt} - 1000\times 10 = 1000\times 20$$

$$\frac{dm}{dt} = 60\,\text{kg/s}$$

9. (*c*) Along horizontal direction,

$M \times 10 \cos30^\circ + 2M \times 5 \cos45^\circ$

$= 2M \times v_1 \cos 30^\circ + M\, v_2 \cos 45^\circ$

$\Rightarrow 5\sqrt{3} + 5\sqrt{2} = 2v_1\frac{\sqrt{3}}{2} + \frac{v_2}{\sqrt{2}}$

Along vertical direction,

$M \times 10 \sin 30^\circ - 2M \times 5 \sin 45^\circ = M\, v_2 \sin45^\circ - 2M\, v_1 \sin30^\circ$

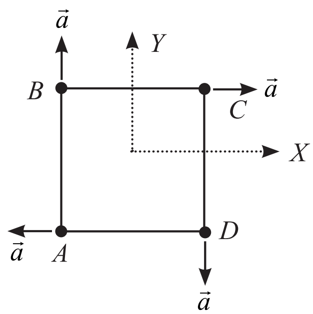
$\Rightarrow 5 - 5\sqrt{2}$

$= \frac{v_2}{\sqrt{2}} - v_1$

Solving $v_1 = \frac{17.5}{2.7} = 6.5$ m/s and $v_2 \approx 6.3$ m/s

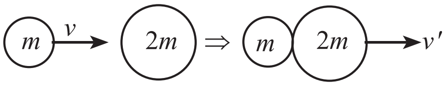
10. (*a*) $\vec{a}_{cm} = \frac{m_A\vec{a}_A + m_B\vec{a}_B + m_C\vec{a}_C + m_D\vec{a}_D}{m_A + m_B + m_C + m_D}$

$$\vec{a}_{cm} = \frac{-ma\hat{i} + 2m\hat{j} + 3ma\hat{i} - 4ma\hat{j}}{10m}$$



$\vec{a}_{cm} = \frac{a}{5}\hat{i} - \frac{a}{5}\hat{j} = \frac{a}{5}(\hat{i} - \hat{j})$

11. (*c*)



Using conservation of linear momentum

⇒ Initial momentum = Final Momentum

$mv + 2m \times 0 = (m + 2m)v'$

$\therefore\ mv = 3mv'$

$v' = \frac{v}{3}$

12. (*a*) Given that,

$v_2 = \frac{n}{m^2} v_2$

$\Rightarrow \frac{L_2}{T_2} = \frac{n}{m^2}\frac{L_1}{T_1}$ ...(*i*)

Also, $a_2 = a_1/mn$

$\Rightarrow \frac{L_2}{T_2^2} = \frac{L_1}{T_1^2} \times \frac{1}{mn}$

$\Rightarrow \left(\frac{T_1}{T_2}\right)^2 = \frac{L_1}{L_2} \times \frac{1}{mn}$

On Putting the volue of $\frac{L_1}{L_2}$ from (*i*)

$\left(\frac{T_1}{T_2}\right)^2 = \frac{T_1}{T_2} \times \frac{m^2}{n} \times \frac{1}{mn}$

$\Rightarrow \frac{T_1}{T_2} = \frac{m}{n^2}$

$\Rightarrow \frac{n^2}{m} T_1 = T_2$ ...(*ii*)

Solving (*i*) and (*ii*), we get

$L_2 = \frac{n^3}{m^3} L_1$

13. (*c*) For position of COM to remain unchanged,

$m_1 x_1 = m_2 x_2$

$\Rightarrow 10 \times 6 = 30 \times x_2$

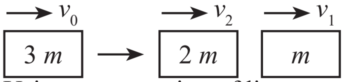
$\Rightarrow x_2 = 2$ cm towards 10 kg block.

14. (*d*) The position vector centre of mass

$\vec{r} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2}{m_1 + m_2}$

$\vec{r} = \frac{1(\hat{i} + 2\hat{j} + \hat{k}) + 3(-3\hat{i} - 2\hat{j} + \hat{k})}{1+3} = -2\hat{i} - \hat{j} + \hat{k}$

15. (*d*)



Using conservation of linear momentum:

$3\ mv_0 = 2\ mv_2 + mv_1$

$\Rightarrow 40 \times 3\ m = 60 \times m + v_2 \times 2\ m$

$\Rightarrow v_2 = 30$ m/s

$KE_i = \frac{1}{2} \times 3m \times (40)^2$

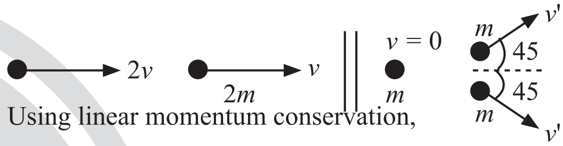
$KE_f = \frac{1}{2} \times m \times (60)^2 + \frac{1}{2} \times 2m \times (30)^2$

$\Rightarrow \frac{KE_f}{KE_i} = \frac{54}{48}$

Hence, Fractional change in kinetic energy

$= 1 - \frac{KE_f}{KE_i} = \frac{1}{8}$

16. (*b*)



Using linear momentum conservation,

$m2v + 2mv = m \times 0 + m\frac{v'}{\sqrt{2}} \times 2$

$4mv = \frac{2mv'}{\sqrt{2}} \Rightarrow v' = 2\sqrt{2}v$

17. (*b*) Using momentum conservation.

$4 \times 10^{-3}(50 - v) - 4v = 0$

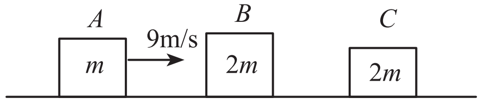
$v = 0.05$ m/s

Impule, $J = mv = 4 \times 0.05$

$J = 0.2$ kg m/s

18. (*d*) Before collision, $u_A = 9$ m/s, $u_B = 0$

Let after collision velocities of A and B be v_A and v_B



Applying conservation of momentum,

$m \times 9 + 2\ mx = m \times v_A + 2mv_B$

$9 = v_A + 2v_B$ ...(*i*)

again, $e = \frac{v_B - v_A}{u_A - u_B}$

Since, collision is elastic, therefore $e = 1$

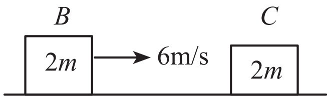
$1 = \frac{v_B - v_A}{9 - 0}$

$9 = v_B - 2v_A$ ...(*ii*)

Solving (*i*) and (*ii*)

$v_B = 6$ m/s and $v_A = -3$ m/s

Now, applying conservation of momentum between B & C



$2m \times 6 + 0 = (2m + 2m)v_C$

$3 \text{ m/s} = v_C$

19. (*a*) Applying the impulse momentum theorem

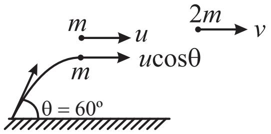
For ball (*a*): $I_1 = \Delta p_1 = 2mu$

Change in momentum,

For ball (*b*): $I_2 = \Delta p_2 = 2mu \cos 45°$

$$\frac{I_1}{I_2} = \frac{2mu}{2mu \cos 45^\circ} = \sqrt{2}$$

20. (*a*)



$p_i = p_f$

$mu + mu\cos\theta = 2mv$

$$\Rightarrow v = \frac{u(1+\cos 60°)}{2} = \frac{3}{4}u$$

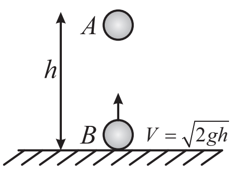
So horizontal range after collision

$$= v\sqrt{\frac{2H_{max}}{g}}$$

$$= \frac{3}{4}u\sqrt{\frac{2u^2\sin^2(60^\circ)}{2g^2}}$$

$$= \frac{3}{4}u^2\frac{\sqrt{\frac{3}{4}}}{g} = \frac{3\sqrt{3}u^2}{8g}$$

21. (*d*)



Time for collision $t_1 = \dfrac{h}{\sqrt{2gh}}$

After t_1, $V_A = 0 - gt_1 = -\sqrt{\dfrac{gh}{2}}$

and $V_B = \sqrt{2gh} - gt_1 = \sqrt{gh}\left[\sqrt{2} - \dfrac{1}{\sqrt{2}}\right]$

At the time of collision $\vec{P}_i = \vec{P}_f$

$$\Rightarrow m\vec{V}_A + m\vec{V}_B = 2m\vec{V}_f$$

$$\Rightarrow -\sqrt{\frac{gh}{2}} + \sqrt{gh}\left[\sqrt{2} - \frac{1}{\sqrt{2}}\right] = 2\vec{V}_f$$

Thus, $V_f = 0$

And height from ground $= h - \dfrac{1}{2}gt_1^2 = h - \dfrac{h}{4} = \dfrac{3h}{4}$

So time $= \sqrt{2 \times \dfrac{\left(\frac{3h}{4}\right)}{g}} = \sqrt{\dfrac{3h}{2g}}$

22. [8] Linear mass density, $\lambda = \lambda_0\left(1 - \dfrac{x^2}{L^2}\right)$ kg/m

$$m = \int \lambda dx = \int_0^L \lambda_0\left(1 - \frac{x^2}{L^2}\right)dx$$

$$= \lambda_0\left(L - \frac{L^3}{3L^2}\right) = \frac{2L\lambda_0}{3}$$

$$X_C = \frac{1}{m}\int x\,\mathrm{dm}$$

$$= \frac{1}{m}\int_0^L x\lambda_0\left(1 - \frac{x^2}{L^2}\right)dx$$

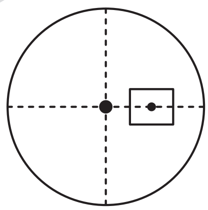
$$= \frac{\lambda_0}{m}\left[\frac{x^2}{2} - \frac{x^4}{4L^2}\right]_0^L = \frac{3}{8}L$$

As, $X_C = \dfrac{3L}{\alpha} = \dfrac{3}{8}L \Rightarrow \alpha = 8$

23. [4] $x_{cm} = y_{cm} = \dfrac{4a}{3\pi}$

$\therefore x = 4$

24. [23]



Let the density is σ

Then original mass $m_0 = \pi a^2\sigma$

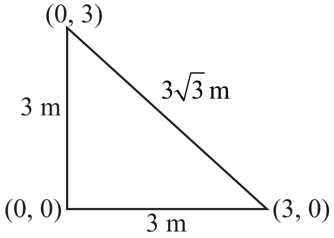
Remaining mass $m' = \left(\pi a^2 - \dfrac{a^2}{4}\right)\sigma = a^2\left(\dfrac{4\pi - 1}{4}\right)\sigma$

Removed mass $m = \dfrac{a^2}{4}.\sigma$

$$\therefore \pi a^2\sigma(0) = \frac{a^2}{4}.\sigma \times \frac{a}{2} + \frac{\sigma a^2}{4}(4\pi - 1)r$$

$$\Rightarrow r = \frac{-a}{2(4\pi - 1)} = \frac{-a}{23.13} = \frac{-a}{23}$$

25. [2]



$$\vec{r}_{\text{com}} = \frac{m(0\hat{i} + 0\hat{j}) + m(3\hat{i}) + m(3\hat{j})}{3m}$$

$$\vec{r}_{\text{com}} = \hat{i} + \hat{j}$$

$$|\vec{r}_{\text{com}}| = \sqrt{2} = \sqrt{x} \Rightarrow x = 2$$

26. [20] Apply conservation of momentum along y-axis

$10\, v_1 = 10\, v_2 \sin 30°$

$\Rightarrow v_2 = 20$ m/s.

27.[400] Velocity of bob after collision,

$$v_1 = \sqrt{5gR} = \sqrt{5 \times 10 \times 0.5} = 5\text{m/s}$$

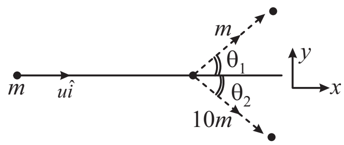
From conservation of momentum,

$m_1 v = m_2 \times 5 - m_1 \times 100$

$$\frac{10}{1000} \times v = \frac{-10}{1000} \times 100 + 1 \times 5$$

$\therefore\ v = 400$m/s

28. [10]



Since energy of m reduced by half

$$\therefore\ u_1 = \frac{u}{\sqrt{2}} \text{ and } \frac{1}{2} \times 10 m v_1^2 = \frac{1}{2}\left(\frac{1}{2} m u^2\right) \Rightarrow v_1 = \frac{u}{\sqrt{20}}$$

Now, momentum in y direction will remain conserved.

$\therefore\ m u_1 \sin\theta_1 = 10 m v_1 \sin\theta_2$

$$\Rightarrow \frac{u}{\sqrt{2}} \sin\theta_1 = 10 \frac{u}{\sqrt{20}} \sin\theta_2 \Rightarrow \sin\theta_1 = \sqrt{10} \sin\theta_2$$